Containerisation: Why, What and How

Containerisation involves encapsulating an application in a container with its own operating environment. This is a short overview on containers and how they are used for application containerisation.

Containers – I am sure all of us have used them in our day-to-day lives to store some things. We usually store something inside a container so that it remains confined within a specified space, is shielded from any external harm or disturbances, and can be kept safe for long. We also use containers when we have to transport anything safely from one place to another. What if we use a container to store or encapsulate an application? Yes, that's what we use containers for in the IT arena.

A container is basically used to store the entire runtime environment of an application, including all its dependencies like the different libraries and configuration files needed to successfully run it. All of these are bundled together into a single package. This ensures that any process inside the container cannot see any other process or resource outside the container.

For long, the hypervisor based virtualisation technology was used to emulate hardware, due to which we could run any OS on top of any other, Linux on Windows, or the other way around. With this technology, both the guest OS and the host OS run with their own kernel. The communication between the guest system and the actual hardware is done with the help of an abstract layer of the hypervisor. This approach actually provides a high level of isolation as well as security, since all communication between the guest and the host is through the hypervisor. But, this approach is much slower and incurs performance overhead due to hardware emulation. Hence, to reduce this overhead, container virtualisation or containerisation was introduced, which allows users to run multiple, isolated user space instances using the same kernel.

Containerisation helps us run any software reliably when it is moved from one computing environment to another. It is really helpful when moving an application that has been developed, from a developer's desktop to a certification or test environment, and further into production. There are cases when we encounter some unexpected exceptions whenever an application or software is migrated from one environment to another, even when that application works fine in the earlier one. Such issues are also observed when software is moved from a physical machine in a data centre to another virtual machine present in a public or private cloud. If we have the application encapsulated in software containers, then we can get rid of all such issues.

Software containers are really helpful when the supporting environment or platform on which the software has been developed is not identical to the one in which it's implemented. Nowadays, containers are also being used to manage different issues caused due to differences in network topology or different security and storage policies. Hence, containers are designed to solve different application